Automated Color-Based Product Sorting Conveyor with 4-DOF Robotic Arm

Group 12

E/21/029 E/21/063 E/21/260

Introduction

In modern industrial manufacturing, the demand for high-efficiency autonomous sorting systems is increasing to replace manual labor and reduce human error. Traditional sorting methods often involve fixed mechanical barriers which lack the flexibility required for diverse product lines. This project focuses on the design and implementation of an automated color-based product sorting conveyor integrated with a 4-degree of freedom (4-DOF) robotic arm. By utilizing an ESP32 microcontroller and sensor-based decision logic, the system provides a practical solution for real-time material handling and sorting.

Objectives

- To design a continuous conveyor belt system for automated product transport.
- To implement real-time color detection using a TCS3200 sensor.
- To develop a 4-DOF robotic arm capable of pick-and-place operations.
- To integrate inverse kinematics for accurate arm positioning.
- To ensure synchronized operation between the conveyor and robotic arm.

Methodology

An infrared (IR) proximity sensor detects the arrival of an object at the picking station and signals the micro controller to stop the conveyor motor. The TCS3200 color sensor then measures the RGB frequency components of the object. The ATmega328P processes the sensor data and determines the object color using predefined thresholds. Based on the detected color, the required joint angles for the 4-DOF robotic arm are calculated. The arm picks the object, places it into the appropriate sorting bin, and then returns to its home position. After completing the operation, the conveyor resumes movement.

Applications

- Industrial packaging and quality control.
- Small-scale pharmaceutical sorting systems.
- Automated recycling and waste segregation.
- Logistics and warehouse automation.

Conclusion

The proposed sorting system demonstrates the effective integration of computer-controlled mechanical actuators with digital sensing. By utilizing the processing capabilities of the ATmega328P, the project provides a robust framework for handling complex motion profiles and sensor fusion. This system highlights the practical application of embedded systems in industrial automation, offering a scalable solution that can be further enhanced with computer vision or IoT integration.